Need help? Should you have any issues with this setup, please email **support@cerelog.com**.

Having Connection Issues? Start Here

If you are having trouble getting the board to connect or stream data, check our dedicated troubleshooting repository for common fixes and debug steps:

View Troubleshooting & Debug Guide

Cerelog BCI/EEG Board User Guide

Welcome! This guide contains everything you need to set up, configure, and use your Cerelog 8-channel Brain Computer Interface board.

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1. Critical Safety & Legal Notices

⚠️ ELECTRICAL SHOCK HAZARD - READ IMMEDIATELY

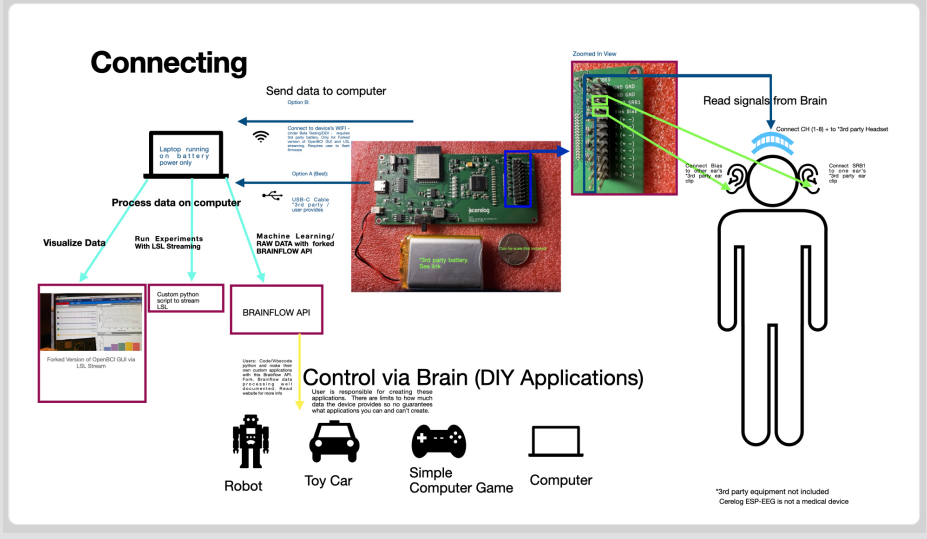
This product **does not have medical-grade isolation circuitry** and is not protected from mains faults or power surges. To prevent risk of serious injury or death from electrical shock, you **MUST** adhere to the following rules:

- **NEVER** use this device on a human subject while it is connected to a desktop computer plugged into a wall outlet.
- **NEVER** use this device on a human subject with a laptop computer that is actively charging from a wall outlet.
- **ALWAYS** operate this device for human data collection using a laptop running on its own battery power, or by connecting the board to an appropriate LIPO battery for power.

Evaluation Board/Kit Important Notice

> is intended for use for **ENGINEERING DEVELOPMENT, DEMONSTRATION, OR EVALUATION PURPOSES ONLY** and is not considered by Cerelog, Inc. to be a finished end-product fit for general consumer use.
>
> Persons handling the product(s) must have electronics training and observe good engineering practice standards. As such, the goods being provided are not intended to be complete in terms of required design-, marketing-, and/or manufacturing-related protective considerations, including product safety and environmental measures typically found in end products that incorporate such semiconductor components or circuit boards.
>
> This evaluation board/kit does not fall within the scope of the European Union directives regarding electromagnetic compatibility, restricted substances (RoHS), recycling (WEEE), FCC, CE or UL, and therefore may not meet the technical requirements of these directives or other related directives.

2. Prerequisites: What You'll Need



Hardware & Accessories (Not Included)

The following links point to recommended third-party vendors. Cerelog Inc. is not affiliated with these vendors and assumes no liability for their products.

- **Power Source:** A 3.7V LIPO battery is recommended for portable, untethered use. **(Qty: 1, Recommended)**
- **Data Cable:** A high-quality USB-C cable rated for data transfer. **(Qty: 1)**
- EEG Electrode Cap**:** A professional cap for placing electrodes on the scalp. **(Qty: 1)**
- Electrode Gel/Paste**:** Conductive gel for a low-impedance connection. **(Qty: 1)**
- Touch-Proof Adapters**:** Essential for connecting the electrode cap to the board. **(Qty: 1)**

> **IMPORTANT: Adapter & Montage**
>
> - The board's default firmware is configured for **Referential Montage**.
> - This setup uses pins 1-8 (+) as the input electrodes and **SRB1** as the common reference/ground.
> - This configuration typically requires only **ONE (1)** touch-proof adapter to operate.

▶ **DIY Headset Alternatives**
Don't have money to buy an entire headset? Make your own...

- Python (version 3.7 or newer).
- Git for cloning software repositories.
- **For BrainFlow use:**

  Microsoft C++ Build Tools (on Windows) or equivalent build essentials on Mac/Linux.
  CMake.

3. Hardware Overview

Connecting the PCB for Data Collection

⚠ CONNECTION INSTRUCTIONS:
For owners of an ESP-EEG V1 — PLEASE READ!

▲ CLICK HERE TO CLOSE

EEG Setup (Scalp Recording)

- **SRB1 pin —** This is the **negative (–) reference electrode for all 8 channels**. Attach an earclip electrode to one ear. Apply conductive gel under the earclip for a good connection.
- **BIAS pin —** Attach a second earclip electrode to the **other ear**, also with conductive gel.
- **CH1+ through CH8+ —** Connect these to your headset electrodes of choice. These are the only channel pins you need for a standard EEG setup.
- **CH1– through CH8– —** Leave these **unconnected**. In the default firmware configuration these pins are not tied to SRB1 and are not used. Only connect them if you change the montage by editing and reflashing the firmware.

Tip: If you run into signal quality issues, give the **GND pin** a try in place of the BIAS pin on the second earclip — it may help.

Not Using All 8 Channels?

For best results, all 8 channels should be in use. If you are using fewer than 8 channels, you should either **tie the unused CH+ pins to one of the active CH+ pins** (connect them together physically), or **disable the unused channels in the firmware** and reflash the board. If unused channels are left floating (unconnected and enabled), the BIAS pin will be less effective at eliminating noise, resulting in poorer signal quality across all channels.

Is your session starting out flat? Try to start the session with the GND pin connected to your earclip and then swap over to the BIAS pin once the session is running. This can help with feedback loop stabilization.

EMG Setup (Muscle Recording)

To record one channel of EMG data (e.g. forearm muscles):

- **CH1+ —** Place on the **top of the forearm** (over the muscle belly).
- **SRB1 —** Place on the **bottom of the forearm** as the negative (–) reference.
- **GND pin —** Place on the **elbow**. For EMG, the GND pin works best as the ground reference rather than the BIAS pin.

Tip: If you experience signal issues during any recording, try swapping the BIAS pin connection for the **GND pin —** it can often provide better performance when signal collection problems arise.

should either **tie the unused CH+ pins to one of the active CH+ pins** (connect them together physically), or **disable the unused channels in the firmware** and reflash the board.

effective at eliminating noise, resulting in poorer signal quality across all channels.

ESP-EEG V1 (8-Channel) — the instructions above apply to this board.

⚠ CONNECTION INSTRUCTIONS:

For owners of an ESP-EEG V2 & 16-Channel — PLEASE READ!

▲ CLICK HERE TO CLOSE

The board supports both a **referential montage (SRB1)** and a **differential montage (CH+ / CH–)**, and you choose between them with the **physical slider switch(es) on the PCB** — not in software.

Choose Your Path: Set the Switch, Then Wire It

The montage is selected by the **slider switch on the PCB**, and the switch position determines how you wire your electrodes. **Pick one of the two paths below and follow it end to end** — they are mutually exclusive.

Tip: The montage switches are **deliberately stiff and hard to move with your fingers**. Use a sewing pin or the tip of a pencil and slide them slowly and carefully into position — do not force them. **16-Channel boards** have **two montage switch banks on the back of the board** instead of one; both must be set to the same mode, whichever path you take.

Path A — Referential Montage (SRB1)

Best for: scalp EEG and most EMG. This is the standard mode.

1. Set the switch: move **all switches to the right**. This puts the board in **SRB1 referential montage mode**. On 16-Channel boards, set both switch banks to the right.

2. Wire your electrodes: in this mode **SRB1** is the shared negative (–) reference for every channel, and **BIAS** drives noise cancellation. A two-channel example:

- **CH1+ and CH2+** — connect to your two recording sites (scalp electrodes for EEG, or over the muscle belly for EMG).
- **SRB1** — connect to one earclip electrode, with conductive gel. This is the negative (–) reference shared by both channels.
- **BIAS** — connect to the earclip on the other ear, also with conductive gel.
- **CH1– through CH8–** (CH16– on 16-Channel boards) — leave these **unconnected** in referential mode.

The same pattern scales to any number of channels: every CH+ you use goes to a recording site, SRB1 to one earclip, BIAS to the other.

> **CH1 is the minimum electrode that must be attached to the user.** You cannot, for example, record three channels using only channels 2, 3 and 4 — one of the channels you use must physically be **CH1**. The BIAS pin stabilises the signal and removes noise through a **feedback loop that runs through Channel 1**, so without CH1 connected the BIAS circuit has nothing to work against.
>
> For the same reason, **a poor CH1 connection degrades BIAS performance across every channel**. Give CH1 the best, lowest-impedance contact of all your electrodes — plenty of conductive gel and a firm, stable placement.

Path B — Differential Montage (CH+ / CH–)

Best for: measuring between two specific points, such as a single muscle in EMG.

1. Set the switch: move the switches to the **opposite position** (not to the right). Each channel then measures between its own **CH+** and **CH–** pins instead of against the shared SRB1 reference.

- **CH1+ and CH1–** — connect this pair across the two points you want to measure between. Repeat per channel: CH2+ / CH2–, CH3+ / CH3–, and so on.
- **SRB1** — not used as the reference in this mode.
- **BIAS** — still connect it (see the BIAS pin note below); noise cancellation works the same way in both modes.

> **CH1 is the minimum electrode that must be attached to the user.** You cannot, for example, record three channels using only channels 2, 3 and 4 — one of the channels you use must physically be **CH1**. The BIAS pin stabilises the signal and removes noise through a **feedback loop that runs through Channel 1**, so without CH1 connected the BIAS circuit has nothing to work against.
>
> For the same reason, **a poor CH1 connection degrades BIAS performance across every channel**. Give CH1 the best, lowest-impedance contact of all your electrodes — plenty of conductive gel and a firm, stable placement.

One Register to Leave As-Is: SRB1 Enable

Montage selection on V2 and 16-Channel boards is handled by the slider switch on the PCB, so the **SRB1 enable bits are set in firmware to work with that switch** and should be left at their default value. If you want to switch between referential and differential mode, move the switch rather than changing this register — toggling SRB1 enable in firmware will not select a montage on these boards. The rest of the firmware is yours to modify as usual.

A Note on the BIAS Pin: BIAS_EN Header & Alternative Bias Options

- **BIAS_EN header** — must be **in place (jumper installed)** for the BIAS pin to work. This is the normal configuration for EEG and for most EMG.
- **ALT_B pin** — for some niche EMG applications, ALT_B may work as an alternative bias pin. It can **only** be used with the **BIAS_EN header removed**.
- **GND pin** — a third option worth trying for EMG. Expect **worse performance** than BIAS or ALT_B; its purpose is to keep the subject near the centre of the supply so readings stay within the board's rail limits.

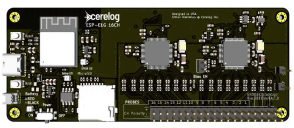
Optional: AVDD & AVSS Header Pins (Powering Active Electrodes)

The board exposes **AVDD** and **AVSS** header pins, which connect to the ADC supply's **+2.5 V and –2.5 V** inputs. If you are using ThinkPulse active electrodes, you can power them from the board's own supplies through these pins. *Cerelog is not affiliated with ThinkPulse.*

16-Channel Boards

Every rule above applies identically — there are simply **16 channels instead of 8**, and **two montage switch banks on the back of the board** rather than one.

ESP-EEG V2 (8-Channel)



ESP-EEG 16-Channel

RECORDING TO MICROSD:
Using the SD card port on ESP-EEG V2 & 16-Channel

The **ESP-EEG V2** and **16-Channel** boards include an **onboard microSD card slot**, which the original V1 board does not have. It lets you record a session directly to a card on the board instead of streaming to a computer.

Insert the Card Before You Power On

The microSD card **must already be inserted before you turn the device on with the power switch**. The card is detected at start-up only — inserting it after the board is powered will not start a recording.

Choosing a Card

- **Capacity:** most cards of **32 GB or less** will work, provided they are **FAT32 formatted**. Cards larger than 32 GB, or cards using a different file system, may not work.
- **Wrong format or too large?** You can reformat the card with your computer's disk utility — aim for a **FAT32** volume of **32 GB or under**.
- **High-endurance cards are recommended** for long recording runs. They significantly reduce the risk of file corruption or interruption over many hours of continuous writing.

What Gets Recorded

Each run produces **one CSV file** on the card. Every row is one sample, with the following columns:

- **Timestamp** — milliseconds since the log started (not wall-clock time).
- **Status** — the 3-byte status word emitted by the ADS1299 with every sample, written in **hex**. It carries the ADC's sync pattern along with lead-off and GPIO flags.
- **CH1 … CH8** — one column per channel, holding **signed ADC counts** (raw 24-bit conversion results, not volts). On **16-Channel** boards this runs **CH1 … CH16** instead.

So a row reads: `timestamp_ms, status_hex, ch1, ch2, … ch8` (or … ch16).

Converting ADC Counts to Voltage

The channel columns are raw signed ADC counts. To convert them (V_{REF} = 4.5 V, gain = 24 by default):

```
Terminal

scale_factor = (2 * 4.5 / 24) / (2**24)   # = 2.2352e-8 V per count
microvolts = raw_val * scale_factor * 1000000
```

Verify Your Recording Before a Long Run

You are responsible for confirming that data collection is actually in progress. Before committing to a long or overnight session, do a short test run first: power the board with the card inserted, record for a minute, then check that the CSV exists on the card and contains sensible data. It is far better to catch a bad card, a bad format, or a loose electrode in a one-minute test than after eight hours of recording.

Advice on Collecting Clean Data

High-Voltage Spike Fix: If you see large voltage spikes in your data caused by environmental interference, try starting your session with the **GND pin** on the second earclip instead of the BIAS pin. Once the signal stabilizes, swap the earclip back to the **BIAS pin** — the data should return to normal. This helps the board's internal feedback loop settle and reduces environmental ringing before the BIAS-driven noise cancellation takes over.

> others) have hardcoded notch-filter frequencies for 50 Hz and 60 Hz, which covers most countries. However, if you live in a region where the power grid suffers from strong frequency instability or operates at a non-standard mains frequency, you will need to manually edit the frequency variable in whichever test script you are using to match your local mains frequency and effectively remove that noise from your recordings.

- **USB-C Port:** For data streaming, powering the board, and charging a connected LIPO battery.
- **Micro JST 1.25 Plug:** 1.25mm pitch battery connector for an optional 3.7V LIPO battery. The board-side connector is a **Molex** PicoBlade, manufacturer product number **0530480210**.
- **Electrode Header Pins:** Two 10-pin headers for connecting the touch-proof adapters. The SRB1 pin on the ADC is used for referential montage mode (default).
- **Power LED (Green):** Located below the Status LED, this indicates the board is receiving power from USB or a battery.
- **GPIO17 Status LED (Green):** This LED indicates the firmware status:

  SOLID ON: Firmware has finished loading, and the board is ready to connect and stream data.

  OFF: Board is not powered, or firmware has not loaded.
- **Battery Charge Status LED:** Located to the left of the power switch (not labeled on the diagram). This LED indicates the charging status of a connected LIPO battery: **RED** when charging, and **GREEN** when fully charged.
- **Onboard Motor (ESP-EEG V1 only):** The original V1 board has a tactile feedback motor that can be controlled via custom firmware. Note that activating it will introduce noise into EEG signals. The **V2 and 16-Channel boards do not have this motor —** it was removed in the newer revisions.

> **Connecting the Battery Correctly**
>
> Carefully align the JST plug so the **RED** wire connects to the 'RED' (+) marking on the PCB and the **BLACK** wire connects to the 'BLK' (-) marking. To remove the connector, do **not** pull the wires; gently squeeze the plastic plug and wiggle it out of the socket.

4. Software Setup & First Data Stream

This section guides you through installing software to visualize your EEG data.

> **A Note for Beginners**
>
> The following steps require using your computer's **Terminal** (also called a command prompt or shell). This is a text-based way to interact with your computer. We recommend using a modern code editor like VS Code, which has a built-in terminal and makes it easy to view files.
>
> **Basic Terminal Commands**
>
> - `ls`: **Lis**ts the files and folders in your current location.
> - `cd FOLDER_NAME`: **C**hanges **d**irectory into the specified folder.

- Before running any commands to clone a repository, first use `cd` to navigate to a folder you can easily find, like your Desktop or Documents folder.
- On macOS, Python and its package manager `pip` are often installed as `python3` and `pip3` to distinguish them from an older, system-installed version of Python. In this guide, whenever you see `python` or `pip`, use `python3` and `pip3` if you are on a Mac.
- To stop a script that is running in the terminal, press `Ctrl + C` (on Windows/Linux) or `Cmd + C` (on Mac).

Note: Device refuses to connect? Read this helpful troubleshooting guide.

Step 1 (Optional): Install Drivers if Needed

Note: This step applies to **V1 devices only**. V2 and 16-Channel boards do not require driver installation.

Modern operating systems like Windows 10/11 and recent versions of macOS/Linux should recognize the board automatically. **Only follow this step if your computer fails to detect the board or assign it a COM port.**

Install CH340C USB-to-Serial Drivers (V1.3 only)

Note: This step applies to V1 devices only. V2 and 16-Channel boards do not require driver installation.

If needed, install the **CH340C** chipset driver. Follow the SparkFun guide linked below for step-by-step instructions for your operating system.

CH340 Driver Installation Guide (SparkFun)

Also: Linux Users May Need the CH341SER Driver

Note: This step applies to V1 devices only. V2 and 16-Channel boards do not require driver installation.

If the CH340C driver above still doesn't work, try the **CH341SER** driver instead (this is a separate driver from the CH340C linked above). This usually works better for **Linux** and can often fix issues with the board not connecting.

Download CH341SER Driver

Note: If you decide to use the CH341SER driver, you may need to hardcode the port names in your BrainFlow test scripts and LSL linking script. Read here for instructions on hardcoding port names.

Raspberry Pi Users: Driver Setup Required

Note: This step applies to V1 devices only. V2 and 16-Channel boards do not require driver installation.

If you own a **Raspberry Pi**, you will need to complete an additional driver setup step before the board can connect. Follow the guide linked below for full instructions.

Raspberry Pi Driver Setup Guide

Debug Step: Hardcoding Port Names

Note: This may apply to V1, V2 and 16CH devices if they are unable to connect.

Some computers assign unusual or unexpected port names to the device. If automatic detection fails, you will need to hardcode the exact port name into your test script.

Read the manual port connection guide here.

Step 2: Choose and Configure Your Visualization Method

Method A: Using the BrainFlow Library

> ⚠️ **This method applies to V1 and V2 devices only.**
>
> **16CH devices** use the native BrainFlow repository and should follow the special connection instructions located in this guide (scroll down to V16CH section) **16CH click here**.

(Recommended / Advanced) This involves building our custom fork of the BrainFlow library from source. This compilation is a **one-time setup for your computer**. Because the core library is written in C++, it must be compiled specifically for your operating system (Windows, Mac, or Linux). You only need to repeat the build process (Step A.2) if you modify the underlying C++ source code files (.cpp, .h). Creating new Python scripts does not require a rebuild.

A.1: Get the Custom BrainFlow Fork

Our board requires a specific fork of BrainFlow. Clone it and navigate into the new directory:

> **Important: Use This Specific Repository**
>
> This version of BrainFlow contains code specific to the Cerelog board. This code has not yet been merged into the official, main BrainFlow repository. You must use the link provided below.

```
Terminal
git clone https://github.com/shakimiansky/Shared_brainflow-cerelog.git
```

```
Terminal
cd Shared_brainflow-cerelog
```

A.2: Build the Library from Source

This crucial step compiles the C++ core of the library. If you make a mistake, manually delete the `build` folder and start this step over.

> **macOS and Raspberry Pi Users: Install Build Tools First**
>
> **macOS:** Install Homebrew (a package manager) and CMake. Open a terminal and run the following commands one by one:
>
> ```
> Terminal
> /bin/bash -c "$(curl -fsSL
> https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
> ```
>
> ```
> Terminal
> brew install cmake
> ```
>
> **Raspberry Pi (Linux):** Raspberry Pi OS uses `apt` instead of Homebrew. Install CMake by running:
>
> ```
> Terminal
> ```

Compilation Steps:
First, create a new directory named `build` inside the `Shared_brainflow-cerelog` folder and navigate into it.

```
Terminal
mkdir build
```

```
Terminal
cd build
```

Next, run the correct `cmake` command for your operating system to prepare the build files.

For Windows (with Visual Studio 2022):

```
Terminal
cmake .. -G "Visual Studio 17 2022" -A x64
```

For macOS / Linux:

```
Terminal
cmake ..
```

Finally, run the build command. This uses 4 processor cores (`-j4`) for faster compilation; you can adjust this number.

```
Terminal
cmake --build . --config Release --clean-first -j4 -- -k
```

A.3: Install the Python Package

> **Raspberry Pi Users: Create a Virtual Environment First**
>
> Raspberry Pi OS requires you to use a Python virtual environment before installing packages. Run these commands **before** proceeding with the steps below:
>
> ```
> Terminal
> python3 -m venv ~/cerelog-venv
> ```
>
> ```
> Terminal
> source ~/cerelog-venv/bin/activate
> ```
>
> You will need to re-activate the virtual environment with `source ~/cerelog-venv/bin/activate` each time you open a new terminal session.

With the core library built, navigate from the `build` folder back up and into the Python wrapper folder.

```
Terminal
cd ..
```

```
Terminal
```

Install the package in "editable" mode. (Remember to use `pip3` on Mac). This links the package to the source files, so you don't need to reinstall after making changes. This command only needs to be run once to set up your environment.

```
Terminal
pip install -e .
```

A.4: Install Required Python Libraries

Before running any scripts, install the required Python libraries. Run the command below. (Use `pip3` on Mac).

```
Terminal
pip install matplotlib pyserial plotly dash scikit-learn "setuptools<82"
```

> **If the command above fails, install them one at a time:**
>
> ```
> Terminal
> pip install matplotlib
> ```
>
> ```
> Terminal
> pip install pyserial
> ```
>
> ```
> Terminal
> pip install plotly
> ```
>
> ```
> Terminal
> pip install dash
> ```
>
> ```
> Terminal
> pip install scikit-learn
> ```
>
> ```
> Terminal
> pip install "setuptools<82"
> ```

A.5: Helpful Videos if You Get Stuck

🎬 2 Quick Videos Using Cerelog ESP-EEG with Brainflow

These are supplemental only. Use them if you get visually stuck — but the **written guide above is far more detailed** and covers setup steps and command scripts that the videos do not include. Always follow the written steps; treat these videos as a secondary visual reference only.

ESP-EEG with Brainflow

General Product Usage Overview

⚠️ *These videos do not cover all setup command scripts. Always refer to the written guide above for the complete setup procedure.*

👇 **Keep scrolling!** The setup guide continues below — this section is not finished yet.

▼ **Click to Expand: Learning How to Write and Filter BrainFlow Data**

(Only for V1 and V2 devices; 16CH device instructions different)

Once you have BrainFlow installed, you can write your own Python scripts to control the board and acquire data. This section provides a complete learning path, starting from a basic script and moving to advanced real-time filtering.

Part 1: Understanding Data Scaling & Basic Scripting

To write your own applications, we recommend studying the example script below in combination with the `unix_plot.py` test script found in the repository. For in-depth information on functions, consult the official BrainFlow documentation.

1. Basic Example: Stream & Static Plot

Save the code below as a new file, for example `my_plot_test.py`, inside the test folder you used previously: `Shared_brainflow-cerelog/python_package/cerelog_tests`. Then run it from the terminal.

Terminal

```
import time
import numpy as np
import matplotlib.pyplot as plt

from brainflow.board_shim import BoardShim, BrainFlowInputParams, BoardIds, Brai

def main():
    """
    A simple script to connect to a BCI, collect data for a few seconds,
    and then plot the EEG data.
    """
    # Board and acquisition parameters
    board_id = BoardIds.CERELOG_X8_BOARD
    duration_seconds = 10

    # --- 1. SETUP THE BOARD ---
    # BrainFlow uses a params object to configure the board
    params = BrainFlowInputParams()

    # Create the BoardShim object
    board = BoardShim(board_id, params)

    # Use a try/finally block to ensure session is always released
    try:
        # Get board info using static methods
        eeg_channels = BoardShim.get_eeg_channels(board_id)
        timestamp_channel = BoardShim.get_timestamp_channel(board_id)
        sampling_rate = BoardShim.get_sampling_rate(board_id)

        print(f"Connecting to {board.get_board_descr(board_id)['name']}...")
        print(f"EEG Channels: {eeg_channels}")
        print(f"Sampling Rate: {sampling_rate} Hz")

        # Prepare the session (finds the board and establishes connection)
```

```
        # --- 2. ACQUIRE DATA ---
        print(f"\nStarting stream for {duration_seconds} seconds...")
        board.start_stream()
        time.sleep(duration_seconds)

        # Stop the stream and get the data from the internal buffer
        board.stop_stream()
        print("Stream stopped. Getting data...")
        data = board.get_board_data()

        if data.size == 0:
            print("Error: No data was collected.")
            return

        # --- 3. PROCESS AND PLOT DATA ---
        # Get the specific data streams from the data array
        eeg_data = data[eeg_channels]
        timestamps = data[timestamp_channel]

        # Create a time axis starting from 0
        time_axis = timestamps - timestamps[0]

        print("Plotting data...")
        plt.figure(figsize=(15, 8))

        # Plot each EEG channel with a vertical offset for clarity
        offset_value = 150 # µV
        for i, channel_data in enumerate(eeg_data):
            plt.plot(time_axis, channel_data + i * offset_value, label=f'Channel

        # Add titles and labels for clarity
        plt.title(f'{duration_seconds} Seconds of Cerelog EEG Data')
        plt.xlabel('Time (s)')
        plt.ylabel('Voltage (µV) - Channels are offset for clarity')
        plt.legend(loc='upper right')
        plt.grid(True)
        plt.tight_layout()

        # Show the plot
        plt.show()

    except BrainFlowError as e:
        print(f"BrainFlow Error: {e}")
    except Exception as e:
        print(f"An unexpected error occurred: {e}")
    finally:
        # --- 4. CLEAN UP ---
        # Always release the session to free the COM port
        if board.is_prepared():
            print("\nRereleasing session.")
            board.release_session()

if __name__ == "__main__":
    main()
```

How It Works: A High-Level Overview

All BrainFlow Python scripts follow a similar pattern:

1. **Import Libraries:** You always need BoardShim and BrainFlowInputParams. For this example, we also import BoardIds to select our board, time for pausing, and numpy and matplotlib for data handling and plotting.

2. **Initialize and Configure:**
   - `params = BrainFlowInputParams()` creates a configuration object. For the Cerelog board over USB, you don't need to set any special parameters.
   - `board_id = BoardIds.CERELOG_X8_BOARD` selects the specific board you're using from a list of supported devices.
   - `board = BoardShim(board_id, params)` creates the main board object that you'll use to control everything.

3. **Get Board Info:** Before connecting, you can get the board's specifications (like its sampling rate and which rows in the data array will contain EEG data) using static methods like

- `board.prepare_session()` finds the board and opens the connection.
- `board.start_stream()` tells the board to start collecting data into an internal buffer.
- `time.sleep(...)` pauses your script to let data collect in the buffer.
- `board.stop_stream()` stops the collection.

5. **Retrieve and Index Data:**
   - `data = board.get_board_data()` retrieves all the data from the buffer as a 2D NumPy array.
   - The `data` array has a specific structure. Each row represents a different channel. By using the channel arrays you fetched in step 3, you can easily select the correct rows: `eeg_data = data[eeg_channels]` for the EEG data, and `time_data = data[timestamp_channel]` for the timestamps.
   - **For advanced users:** If you wish to hardcode the indices, the data array for the Cerelog board is typically organized as follows, though using the `get...` methods is strongly recommended:
     Row 0: Packet Counter
     Rows 1-8: EEG Channels
     Row 9: Marker Channel
     Row 10: Timestamp Channel

6. **Clean Up:** Always call `board.release_session()` in a `finally` block. This ensures the connection to the board is closed properly, freeing up the COM port for other applications.

> ⚠️ **IMPORTANT: Scaling Cerelog Data in BrainFlow**
>
> Unlike some other devices, the Cerelog board provides raw, unscaled **voltage** data to give you maximum fidelity and control. When processing this data in your own BrainFlow scripts, you **must apply a scaling factor** to convert the raw voltage values into standard units.
>
> Failure to apply the voltage conversion will result in incorrect plots and analysis, with a flat-looking vertical (voltage) scale.
>
> **Required Data Conversions:**
>
> Inside your data processing loop or after you retrieve data with `board.get_board_data()`, apply the following:

```
Terminal
# Get the raw data streams from the main data array
eeg_data_raw = data[eeg_channels]
timestamps_raw = data[timestamp_channel]

# --- 1. Scale EEG Data (Vertical Axis) ---
# The board returns data in Volts (V). To get microvolts (µV),
# the standard unit for EEG, you must multiply by 1,000,000.
eeg_data_microvolts = eeg_data_raw * 1e6

# --- 2. Timestamp Data (Horizontal Axis) ---
# The board sends timestamp data in milliseconds, but BrainFlow
# automatically converts it to Unix time in seconds. No additional
# scaling is needed. Simply make it relative to the start time.
time_axis_seconds = timestamps_raw - timestamps_raw[0]
```

> Always use the scaled variable `eeg_data_microvolts` for your plotting and analysis functions. The timestamp data does not need to be scaled — BrainFlow handles the conversion to seconds automatically.

Part 2: Advanced Real-Time Filtering

Reaching clean EEG data requires more than just connecting to the board. You must apply digital signal processing (DSP) to remove noise and isolate the brainwaves. While the basic test above shows you how to stream, we recommend using the script below (`filtered_plot.py`) as your main reference for creating robust applications.

> **Understanding the Filters**
>
> The code below implements a chain of filters crucial for real-time BCI. Here is what they do:
>
> **1. Detrend:** Removes the DC offset (the constant voltage drift) so the signal centers around 0 microvolts.

```
DataFilter.detrend(eeg_plot_data[i], DetrendOperations.CONSTANT.value)
```

2. Low-Pass (100Hz): Removes high-frequency noise that isn't EEG. We use a 4th-order Butterworth filter for stability in real-time streams.

```
Terminal

DataFilter.perform_lowpass(eeg_plot_data[i], sampling_rate, 100.0, 4,
FilterTypes.BUTTERWORTH, 0)
```

3. Band-Stop (Notch Filters): Specifically targets 50Hz and 60Hz noise caused by electrical mains (wall outlets).

```
Terminal

DataFilter.perform_bandstop(eeg_plot_data[i], sampling_rate, 48, 52, 3,
FilterTypes.BUTTERWORTH, 0)
DataFilter.perform_bandstop(eeg_plot_data[i], sampling_rate, 58, 62, 3,
FilterTypes.BUTTERWORTH, 0)
```

4. High-Pass (0.5Hz): Removes extremely slow drifts caused by movement or sweat.

```
Terminal

DataFilter.perform_highpass(eeg_plot_data[i], sampling_rate, 0.5, 4,
FilterTypes.BUTTERWORTH, 0)
```

5. Rolling Median: A final smoothing pass that helps remove sudden spikes without blurring the signal as much as a mean filter would.

```
Terminal

DataFilter.perform_rolling_filter(eeg_plot_data[i], 3,
AggOperations.MEDIAN.value)
```

2. Advanced Example: Real-Time Filtering & Scrolling

Save the code below as `filtered_plot.py` and run it.

```
Terminal

import time
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation
from brainflow.board_shim import BoardShim, BrainFlowInputParams, BoardIds, Bra
from brainflow.data_filter import DataFilter, FilterTypes
from brainflow.data_filter import NoiseTypes, DetrendOperations, AggOperations,
# --- Configuration ---
BOARD_ID = BoardIds.CERELOG_X8_BOARD
SECONDS_TO_DISPLAY = 10
UPDATE_INTERVAL_MS = 40
Y_AXIS_PADDING_FACTOR = 1.2

# --- Global variables ---
board = None
eeg_channels = []
sampling_rate = 0
window_size = 0
data_buffer = np.array([])
```

A.6: Run Included Tests

(Only for V1 and V2 devices; 16CH device instructions different)

Wait for the Green LED!

Before running any test that connects to the board, make sure you've plugged it in and

The repository includes a comprehensive test suite. To run them, you must be inside the `python_package` directory.

1. Recommended First Test: Visual Data Stream

We strongly recommend running the **Filtered Plot** test first. This script provides a real-time, filtered graph of all 8 channels, allowing you to instantly verify signal quality and electrode contact.

▼ **Raspberry Pi Users: Read This First**

1. Activate Your Virtual Environment

Before running any test scripts, make sure your virtual environment is active. If you opened a new terminal since Step A.3, re-activate it:

```
Terminal
source ~/cerelog-venv/bin/activate
```

2. Hardcode the Serial Port

All BrainFlow test scripts — including `filtered_plot.py` — must have the serial port name hardcoded to run on Raspberry Pi. Automatic port detection does not work on Pi.

In each script, find the line:

```
Terminal
params = BrainFlowInputParams()
```

And add the following line directly after it:

```
Terminal
params.serial_port = "/dev/ttyCH341USB0"
```

Note for V2 and 16CH devices: These boards do not use the CH341 chip, so `/dev/ttyCH341USB0` will not exist. You need to manually find your board's port name in the terminal (for example, run `ls /dev/tty*` before and after plugging in the board and look for the new entry), then use that port name in the line above.

You must make this edit in every BrainFlow test script you want to run on the Pi.

3. Running Test Scripts Over SSH

Test scripts that display plots must be run from the Raspberry Pi's own terminal. If you are connected over SSH, the plot window cannot open on your remote machine by default. To run a script over SSH and have the plot display on the Pi's screen, prefix the command with `DISPLAY=:0`:

```
Terminal
DISPLAY=:0 python filtered_plot.py
```

```
Terminal
cd cerelog_tests
```

```
Terminal
python filtered_plot.py
```

Did you get: `ModuleNotFoundError: No module named 'pkg_resources'`?

This is because your version of **setuptools** has removed this legacy module. To fix it, downgrade setuptools by running:

```
Terminal
python3 -m pip install "setuptools<82"
```

On Windows, use `python` instead of `python3`:

```
Terminal
python -m pip install "setuptools<82"
```

Note: Device refuses to connect? Read this helpful troubleshooting guide.

2. Run PCB Health Check Suite

(Only for V1 and V2 devices; 16CH device instructions different)

To verify the technical integrity of the board (timestamps, packet loss, serial connection), run the automated test suite.

```
Terminal
python run_all_tests.py
```

Key Tests Explained

`test_brainflow.py`

The best starting point for developers. It demonstrates the full cycle: connecting to the board, streaming data, and calculating basic stats. It also saves the raw data to a CSV file.

`unix_plot.py`

Highly recommended test. This script connects to the board, streams data, and provides a real-time plot, making use of the board's Unix timestamps for accurate timing.

`test_serial.py`

A low-level test that bypasses BrainFlow to communicate directly with the board. Use this if you suspect a fundamental serial connection or packet format issue.

`test_unix_timestamps.py`

An advanced test for verifying time synchronization between the board and your PC, crucial for time-sensitive experiments.

For a complete list of tests and advanced debugging options, please see the `README.md` file inside the `cerelog_tests` directory.

A.7: Play BCI Pong (New)

(Only for V1 and V2 devices; 16CH device instructions different)

⚠ EPILEPSY & LIABILITY WARNING

> patterns which can trigger seizures in individuals with photosensitive epilepsy. By choosing to run these games, you acknowledge these risks. You agree to hold Cerelog Inc. and all affiliated parties harmless from any liability or damages that may arise from their use.

Once you have installed the BrainFlow library (Steps A.1 through A.3), you can play the new BrainFlow-enabled Pong game.

> **Performance Requirement**
>
> This game uses SSVEP (Steady-State Visually Evoked Potential) which relies on high-frequency flickering lights. It will **not work properly on slow computers** because the screen cannot refresh fast enough to create the required brain stimulation frequencies.

To play, navigate to the games folder inside the Python package:

```
Terminal

cd Games/"Fancy JS Game"
```

Run the game script (remember to use `python3` on Mac):

```
Terminal

python pong_game_brainflow.py
```

Note: You may find other games in the Games directory that are currently under development.

Method B: OpenBCI GUI or Lab Stream Layer (LSL)

(NEW and Easy) Use with the modified version of the OpenBCI GUI or Lab Stream Layer (LSL) here: https://github.com/Cerelog-ESP-EEG/How-to-use-OpenBCI-GUI-fork

<details>
<summary>▶ Method C: Legacy Python Web Plotter (Deprecated) (V1 devices only)</summary>
This method is outdated and no longer recommended. Click to expand if needed.
</details>

5. Running an EEG Session

1. **Hardware Assembly:** Connect your EEG cap cable(s) to the required touch-proof adapter(s), then plug the adapter(s) into the electrode headers on the Cerelog board.

2. **Prepare the Subject:** Fit the EEG cap on the subject's head. Apply a small amount of conductive gel into each electrode cavity to ensure good skin contact.

3. **Connect Reference & Bias (Ear Clips):** For accurate BCI data collection, you **must** use two ear clip electrodes with conductive gel.
   - Connect the first ear clip to the **BIAS** pin.
   - Connect the second ear clip to the **SRB1** pin (this acts as the Ground/Reference in the default montage).
   - Attach one clip to each earlobe.

4. **Connect and Stream:** Connect the board to your computer via USB-C. Launch your chosen software or script, and start the data stream.

> ⚠️ **Safety Reminder**
>
> As per the notice in Section 1, only connect to a laptop running on its own battery power. **DO NOT** use this device if the laptop is charging or connected to a desktop computer plugged into a wall outlet.

5. **Check Signal Quality:** Observe the incoming data. If a channel is flat or excessively noisy, check the corresponding electrode's position and consider re-applying a small amount of gel.

> **IMPORTANT: Signal Quality & Artifacts**
>
> The human body acts like an antenna, picking up the 50/60 Hz hum from nearby electrical mains. This noise is often much stronger than the tiny EEG signals from the brain. Getting a clean signal is critical for any BCI application.
>
> **1. Effective Filtering & "Jaw Clench" Artifacts**
>
> The Cerelog board utilizes highly effective filtering on the BIAS pin to cancel out common-mode noise. Because of this superior noise cancellation, you might not see massive, screen-filling spikes when you clench your jaw, unlike on some cheaper or older hardware. **This is normal.** It means the board is successfully filtering out the noise generated by the muscle movement, allowing the EEG signal to remain cleaner.
>
> **2. Use the Bias Probe Correctly**
>
> The **BIAS probe** is the single most important tool for noise cancellation. It measures the mains hum from the body and subtracts it from the EEG channels. **You will get poor data without it.** Ensure the ear clips used for BIAS and SRB1 have plenty of conductive gel.
>
> **3. Filter Your Data**
>
> For best results, especially with BrainFlow, we highly recommend applying a **notch filter** to your data in software to remove any remaining 50 Hz or 60 Hz noise.

6. Advanced Usage: Firmware Customization

> **Firmware Access is Restricted**
>
> To receive the firmware source code or if you need the code to change montages **(V1 devices only)**, please email **support@cerelog.com** with your proof of purchase (e.g., order number). Firmware can be modified and flashed using the Arduino IDE.
>
> **V2 and 16CH devices:** you do not need firmware changes to change the montage. Change it with the montage switches at the bottom of the PCB, as noted in the **connection info**.

> ⚠️ **WARNING: Custom Firmware & Software Compatibility**
>
> Any modifications you make to the firmware are **not guaranteed to work** with the provided BrainFlow fork, Python plotter, or games. Changes to critical parameters like sample rate, data resolution, or packet structure **will** break the software.
>
> If you modify the firmware, you may be required to make heavy modifications to the software to maintain compatibility. This is an advanced task and is your own responsibility.

> **V1 Flash Instructions: Configuring the Arduino IDE**

- **Board:** Navigate to `Tools > Board > ESP32 Arduino` and select **'ESP32 WROOM DA Module'**.
- **Port:** Navigate to `Tools > Port` and select the COM port corresponding to your Cerelog board.

Note: V2 and 16CH devices have their own special flash instructions. Read them in the firmware folder.

Modifying the firmware is necessary to:

- **Change Electrode Montage (V1 devices only):** Switch from the default **Referential** mode to **Sequential** mode. Sequential mode measures the difference between P and N pins directly, requiring two adapters. *V2 and 16CH devices change the montage with the montage switches at the bottom of the PCB, as noted in the **connection info**; no firmware change is needed.*
- **Adjust Gain:** Change the Programmable Gain Amplifier (PGA) setting (1, 2, 4, 6, 8, 12, or 24).
- **Control Peripherals:** Write custom code to use the debug LEDs, or—on **V1 boards only**—the haptic feedback motor. The V2 and 16-Channel boards do not have a motor.
- **Change Sample Rate:** This is a highly advanced modification. Altering the sample rate or resolution will change the data stream format and will require you to rewrite the data parsing logic in your software.

CRITICAL: Procedure After Flashing Firmware

1. After a successful flash, you **MUST** perform a full power cycle. Unplug the USB-C cable. If a battery is connected, unplug it as well.
2. Reconnect the power (USB-C and/or battery).
3. Your computer may assign the board a **new COM port number**. Check your Device Manager to see the new port assignment.
4. When you run your software again, both the Python plotter and a properly configured BrainFlow script should find the new port automatically. If you have hard-coded a specific COM port in a custom script, you will need to update it manually.

7. Troubleshooting Guide

Problem: Board not detected (no COM port appears).

- First, try a different USB port and ensure you are using a **USB-C data cable**, not a charge-only one.
- If it's still not detected, proceed with the optional CH340C driver installation in Section 4.
- Check for the green power LED on the board. If it's not on, there is a power issue.

Problem: My script fails to connect.

- Make sure the green status LED is on before running your script. This indicates the firmware is loaded and ready.
- Try a full power cycle. Disconnect the USB-C cable, wait a few seconds, and plug it back in before re-running your script.
- Make sure no other program is already connected to the board's COM port. Only one application can use the port at a time.
- If you recently flashed new firmware, see the "Procedure After Flashing" notes in Section 6.

Problem: Data is extremely noisy or flat.

- Poor electrode connections are the most common cause of bad signal quality. Ensure that all electrodes, especially the **BIAS probe**, have a secure, low-impedance connection.
- Ensure you are not near sources of high electromagnetic interference (e.g., power bricks, fluorescent lights, powerful fans).
- Re-apply conductive gel to the problematic electrodes.

8. Product Demonstrations

See the Cerelog ESP-EEG board in action. These demos showcase the real-time data streaming and BCI control capabilities.

Product Demos

See the Cerelog ESP-EEG board in action. These demos showcase the real-time data streaming and BCI control capabilities.

Product Overview Guide and Live Data Visualization

Video demonstrating how to use product for real-time EEG signal data streaming and plotting.

9. Applications & Possibilities

This board is a versatile tool for developers and researchers. With custom software, you can explore a wide range of applications.

- **Multimodal Signal Acquisition:** Capture data for EEG (brain), EMG (muscle), ECG (heart), and EOG (eye movement).
- **Neuroscience Paradigms:** Design experiments based on P300, Event-Related Potentials (ERP), and Steady-State Visually Evoked Potentials (SSVEP).
- **BCI & Control Systems:** Develop systems to quantify cognitive or emotional states, or to control robotics and other external devices.

10. Appendix: Technical Details

Technical Specifications

Input Channels	8 differential channels + 1 bias probe input
ADC	Texas Instruments ADS1299
ADC Resolution	24-bit
Sample Rate	250 SPS (default, firmware configurable)

Connectivity	USB-C (data/power), plus Bluetooth Low Energy (BLE) and WiFi wireless streaming.

Understanding Electrode Montages

The board can be configured in two primary ways:

- **Referential Montage (Default):** Each channel measures the potential difference between a positive input (Pins 1-8+) and a common reference. In this mode, the **SRB1 pin** is configured as the common ground/reference for all channels. This configuration is standard for most BCI applications and typically requires only a **single 10-pin adapter**.
- **Sequential / Bipolar Montage (Requires Firmware Mod on V1 devices only):** Each channel measures the potential difference between a positive input (INxP) and a negative input (INxN). For example, Channel 1 is (IN1P - IN1N). This requires **two 10-pin adapters** to connect all necessary pins. On **V2 and 16CH devices**, select this montage with the montage switches at the bottom of the PCB instead, as noted in the **connection info**.

> ### A Note on Performance
>
> The core of this board is the research-grade TI ADS1299 ADC. While the chip has exceptional theoretical performance (capable of resolving ~10µV signals), real-world board performance will be influenced by your setup, environmental noise, and electrode contact quality. For complete details on the ADC, see the official TI ADS1299 Datasheet.

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